

# Frontend Engineering Fundamentals: A 6-Week Intensive Course

## Course Overview

This curriculum transforms complete beginners into job-ready frontend developers through structured learning, hands-on practice, and a comprehensive capstone project. Each week builds progressively on previous knowledge, culminating in a portfolio-worthy final project.

## WEEK 3: Modern JavaScript & CSS Grid

### Learning Objectives

By the end of this week, students will:

- Master ES6+ features (arrow functions, destructuring, spread/rest, modules)
- Understand array methods (map, filter, reduce, forEach)
- Build complex layouts with CSS Grid
- Work with forms and form validation
- Implement client-side routing concepts

### Learning Materials

#### ES6+ JavaScript

##### Videos:

- "Modern JavaScript From The Beginning" by Traversy Media (ES6 sections, 2 hours)  
<https://www.youtube.com/watch?v=BI1o2H9z9fo>
- "JavaScript Array Methods" by Web Dev Simplified (30 mins)  
<https://www.youtube.com/watch?v=R8rmfD9Y5-c>
- "Destructuring in JavaScript" by Fireship (8 mins)  
<https://www.youtube.com/watch?v=UgEaJBz3bjY>

- "JavaScript Modules" by Web Dev Simplified  
<https://www.youtube.com/watch?v=cRHQNNcYf6s>

### **Reading:**

- JavaScript.info - Advanced Working with Functions  
<https://javascript.info/advanced-functions>
- JavaScript.info - Modules  
<https://javascript.info/modules>
- MDN - ES6 Features  
<https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide>

**Practice:** Refactor Week 2 project using ES6+ features

### **CSS Grid Mastery**

#### **Videos:**

- "CSS Grid Tutorial" by Wes Bos (free course, 4 hours)  
<https://cssgrid.io/>
- "Grid vs Flexbox - When to use which?" by Kevin Powell (20 mins)  
<https://www.youtube.com/watch?v=3elGSZSWTbM>
- "Learn CSS Grid in 20 Minutes" by Web Dev Simplified  
<https://www.youtube.com/watch?v=9zBsdzdE4sM>

#### **Reading:**

- CSS-Tricks - A Complete Guide to CSS Grid  
<https://css-tricks.com/snippets/css/complete-guide-grid/>

- MDN - CSS Grid Layout

[https://developer.mozilla.org/en-US/docs/Web/CSS/CSS\\_Grid\\_Layout](https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_Grid_Layout)

**Practice:** Build complex layouts (magazine style, dashboard)

## Forms & Validation

### Videos:

- "Form Validation with JavaScript" by Traversy Media (40 mins)  
<https://www.youtube.com/watch?v=rsd4FNGTRBw>
- "Regular Expressions (RegEx) Tutorial" by Web Dev Simplified (30 mins)  
<https://www.youtube.com/watch?v=rhzKDrUiJVk>

### Reading:

- MDN - Client-side Form Validation  
[https://developer.mozilla.org/en-US/docs/Learn/Forms/Form\\_validation](https://developer.mozilla.org/en-US/docs/Learn/Forms/Form_validation)
- JavaScript.info - Forms, Controls  
<https://javascript.info/forms-controls>

**Practice:** Create a multi-step form with validation

## Week 3 Challenge Project

Build a Recipe Finder Application

Requirements:

- Search recipes using a public API ([TheMealDB](#) or [Spoonacular](#))
- Display results in a responsive grid layout
- Show recipe details on click (modal or separate view)
- Filter by category and ingredients
- Save favorite recipes to localStorage
- Form validation for search inputs

- Loading states and error handling

**Deliverable:** Live app + code repository

**Success Criteria:**

- Clean ES6+ JavaScript code
- Proper API integration with error handling
- Advanced CSS Grid layouts
- Form validation with user feedback
- Excellent UX with loading and error states